PHYSIOLOGY · MCQ PRACTICE

Liver Physiology — MCQ Practice

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GIT BLOCK

Physiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Questions

1. A pathologist looking at a liver biopsy identifies a hexagonal unit with a portal triad at each corner. Which vessel lies in the middle of this unit? EASY

A. Portal vein
B. Central vein
C. Hepatic artery
D. Bile duct
E. Hepatic vein

2. A 62-year-old man with complete blockage of the common bile duct passes pale stools. The absence of which substance from the intestine explains the loss of the normal brown colour? EASY

A. Lecithin
B. Cholesterol
C. Cholic acid
D. Stercobilin
E. Glutathione

3. Portal blood constantly carries bacteria from the gut, yet a healthy volunteer's systemic blood stays almost free of them. Which cells are responsible? EASY

A. Hepatic stellate cells in the space of Disse
B. Cholangiocytes lining the bile ducts
C. Kupffer cells in the sinusoids
D. Hepatocytes of the hepatic plates
E. Intestinal I cells

4. A 55-year-old woman with severe chronic liver failure has low levels of most plasma proteins. Which major class of plasma proteins is NOT synthesised by the liver? EASY

A. Albumin
B. Acute-phase proteins
C. Hormone-binding proteins
D. Clotting factors
E. Immunoglobulins

5. A healthy student eats a large meal. Compared with the resting state, how does the supply of blood to her liver change? MEDIUM

A. The portal vein's share rises from 75% to about 90%
B. The hepatic artery's share rises from 25% to about 90%
C. The portal vein's share falls from 75% to about 25%
D. Both vessels come to supply equal shares
E. The shares stay fixed at 25% and 75%

6. Pressures are measured along the hepatic circulation of a healthy adult. Which set matches the lecture (portal vein / hepatic artery / sinusoid / vena cava)? MEDIUM

A. 90 / 9 / 4 / 0 mmHg
B. 4 / 90 / 9 / 0 mmHg
C. 9 / 90 / 20 / 4 mmHg
D. 9 / 90 / 4 / 0 mmHg
E. 0 / 90 / 9 / 4 mmHg

7. In most capillary beds plasma proteins help hold water inside the vessel. Why is this protection lost in the hepatic sinusoids, leaving hydrostatic pressure as the only force controlling fluid movement? MEDIUM

A. Kupffer cells remove plasma proteins from sinusoidal blood
B. Plasma proteins pass freely through the large endothelial openings
C. Hepatocytes stop producing albumin whenever sinusoidal pressure is low
D. Bile salts in portal blood cancel the osmotic effect of proteins
E. Sinusoidal pressure is higher than hepatic arterial pressure

8. In the post-prandial period secretin acts on the biliary system. What effect on bile is expected? MEDIUM

A. The gallbladder contracts and the sphincter of Oddi relaxes
B. Bile salts are reabsorbed from the ileum
C. Bilirubin is conjugated with glucuronic acid
D. Glucose and amino acids are added to it
E. Its water and bicarbonate content increases

9. The bile ductules actively return glucose and amino acids from bile to the circulation. What does the lecture give as the likely benefit? MEDIUM

A. It raises the bicarbonate and water content of bile
B. It makes bilirubin more water-soluble before secretion
C. It limits bacterial overgrowth in stored bile
D. It allows bile salts to emulsify large lipid globules
E. It triggers CCK release from intestinal I cells

10. An ultrasound in a fasting patient shows a full gallbladder. The bile inside it is more concentrated than the bile leaving the liver. Which process explains this? MEDIUM

A. Absorption of water and ions by the gallbladder
B. Secretion of extra bile salts by the gallbladder wall
C. Bicarbonate secretion in response to secretin
D. Hydrolysis of glutathione by gamma-glutamyltranspeptidase
E. Active transport of bilirubin by MRP-2

11. A 40-year-old man has had his terminal ileum surgically removed. Which consequence for bile salt handling is expected? MEDIUM

A. Bile salts can no longer be secreted into the bile canaliculi
B. The gallbladder can no longer concentrate bile by absorbing water
C. CCK can no longer be released when chyme enters the duodenum
D. Fewer bile salts return to the liver, so it must make more
E. Bilirubin can no longer be conjugated inside the hepatocyte

12. A 68-year-old man with cardiac failure and a high right atrial pressure has an enlarged liver. How much extra blood can the liver store in this setting, and where? MEDIUM

A. About 100 mL, in the bile canaliculi of the plates
B. 0.5 to 1 litre, in the hepatic veins and sinuses
C. 0.5 to 1 litre, in the portal lymphatic vessels
D. 3 to 7 litres, in the space of Disse around hepatocytes
E. None, because the liver cannot expand to hold blood

13. A labelled plasma protein leaves a hepatic sinusoid through the endothelial fenestrae. Which compartment does it enter first? MEDIUM

A. The space of Mall
B. The bile canaliculus
C. The cisterna chyli
D. The central vein
E. The space of Disse

14. A newborn has markedly reduced activity of UDP-glucuronosyltransferase. Which step of bilirubin handling is impaired, and where does it normally take place? MEDIUM

A. Binding to albumin, in the circulating blood
B. Uptake into the hepatocyte, through the OATP carrier
C. Conjugation with glucuronic acid, in the smooth ER
D. Active secretion into bile, through the MRP-2 pump
E. Conversion to stercobilin, in the intestinal lumen

15. A 50-year-old woman's portal venous flow falls sharply. Which statement best describes how far the hepatic arterial buffer response protects her liver? HARD

A. Arterial flow rises and stabilises oxygen delivery, but total flow is not fully preserved
B. Arterial flow rises enough to bring total liver blood flow fully back to its resting value
C. Arterial resistance rises, because adenosine is washed out more quickly
D. Arterial flow falls in parallel, so that the two inflow pressures stay matched
E. Arterial flow rises and reliably protects the biliary tree if its arterial supply is lost

16. Two patients have ascites. Patient X has a vena caval pressure of 12 mmHg from heart failure. In patient Y, portal flow through the liver is blocked. Which mechanism applies specifically to patient Y? HARD

A. Hepatic lymph flow rises to 20 times normal and fluid sweats from the surface of the liver capsule
B. Kupffer cells release interleukins that make the peritoneum secrete fluid
C. Hepatic arterial pressure of 90 mmHg is transmitted directly to the liver capsule
D. Bile salts can no longer return from the ileum, so bile leaks into the abdominal cavity
E. High portal capillary pressure causes gut-wall oedema and transudation through the gut serosa

17. After an overnight fast both liver and skeletal muscle still contain glycogen, yet only the liver adds glucose to the blood. What explanation does the lecture give? HARD

A. Muscle glycogen cannot be broken down at all during an overnight fast
B. Only the liver expresses the complete urea cycle needed to release glucose
C. Muscle lacks the cytochrome P450 enzymes that release glucose from glycogen
D. Only the liver has glucose-6-phosphatase; muscle uses its glycogen locally
E. Liver glycogen is released into bile and then reabsorbed into portal blood

18. A 59-year-old man with advanced liver failure becomes confused and has a high blood ammonia level. Which statement about ammonia is correct? HARD

A. It comes mainly from red-cell breakdown, and the kidneys convert it to urea through the urea cycle
B. It comes mainly from the colon and kidneys, and only the liver has the complete urea cycle
C. It comes mainly from muscle metabolism, and Kupffer cells remove it by phagocytosis
D. It comes mainly from the colon and kidneys, and the liver excretes it unchanged in bile
E. It comes mainly from hepatocytes, and it cannot cross the blood–brain barrier

19. Which sequence correctly traces hepatic lymph of the deep system from its formation back to the blood? HARD

A. Sinusoid → space of Mall → space of Disse → portal lymphatics → cisterna chyli → thoracic duct → left venous angle
B. Sinusoid → space of Disse → bile canaliculus → portal lymphatics → cisterna chyli → thoracic duct → left venous angle
C. Sinusoid → space of Disse → space of Mall → portal lymphatics → cisterna chyli → thoracic duct → left venous angle
D. Sinusoid → space of Disse → space of Mall → portal lymphatics → thoracic duct → cisterna chyli → left venous angle
E. Sinusoid → space of Disse → space of Mall → portal lymphatics → cisterna chyli → thoracic duct → right venous angle

20. Two men have chronic liver failure. Patient X has a low platelet count from loss of a hormone made by the liver. Patient Y has features of oestrogen excess. Which pairing of lost liver functions fits? HARD

A. X: thrombopoietin production; Y: catabolism of oestrogen
B. X: blood-cell production, as in the fetus; Y: synthesis of hormone-binding proteins
C. X: storage of vitamin B12 and iron; Y: phase II esterification of bile salts
D. X: destruction of senile cells by Kupffer cells; Y: synthesis of steroids
E. X: synthesis of heparin; Y: synthesis of somatomedin

✓ Answer key

Q	Answer	Level	Why (lecture slide)
1	**B**	EASY	Each hexagonal lobule has a central vein in the middle and portal triads at the periphery. [Slide 8]
2	**D**	EASY	Bilirubin secreted in bile is broken down in the intestine; its product stercobilin gives faeces their brown colour. [Slide 23]
3	**C**	EASY	Phagocytic Kupffer cells filter particulates so that almost no gut bacteria pass from portal blood to the systemic circulation. [Slide 7]
4	**E**	EASY	The only major class of plasma proteins not synthesised by the liver is the immunoglobulins. [Slides 38, 43]
5	**A**	MEDIUM	The usual 25% arterial + 75% portal split shifts to 90% from the portal vein in the post-prandial period. [Slide 15]
6	**D**	MEDIUM	Low-pressure portal vein (9) and high-pressure hepatic artery (90) merge in the porous sinusoids at only 4 mmHg; blood reaches the vena cava at 0 mmHg. [Slide 17]
7	**B**	MEDIUM	Because proteins cross the large openings freely, the protective osmotic pressure is eliminated and hydrostatic pressure alone controls fluid movement. [Slide 18]
8	**E**	MEDIUM	Secretin increases the water and HCO_3^- content of bile; the ductules secrete bicarbonate in response to it. [Slides 22, 27]
9	**C**	MEDIUM	Removing glucose and amino acids is likely important to prevent bacterial overgrowth in bile, particularly during gallbladder storage. [Slide 22]
10	**A**	MEDIUM	Bile fills the gallbladder and is stored; the gallbladder concentrates the bile salts by absorbing water and ions. [Slide 24]
11	**D**	MEDIUM	Bile salts recirculate from the ileum via the portal system; this recirculation reduces the demand to synthesise new bile salt. [Slide 24]
12	**B**	MEDIUM	The liver is an expandable venous reservoir; up to 0.5–1 L of extra blood is stored in the hepatic veins and sinuses when right atrial pressure is high. [Slide 29]
13	**E**	MEDIUM	Hepatic lymph originates from plasma filtered through the fenestrae into the space of Disse, between the sinusoidal endothelial cells and hepatocytes. [Slide 31]
14	**C**	MEDIUM	Bilirubin is conjugated to glucuronic acid by UDP-glucuronosyltransferase, located mainly in the smooth endoplasmic reticulum. [Slide 36]
15	**A**	HARD	Reduced adenosine washout dilates the arterioles and partially compensates; it stabilises oxygen delivery but does not completely preserve total flow or protect the biliary tree. [Slide 16]
16	**E**	HARD	Blocked portal flow raises capillary pressure in the whole portal system → oedema of the gut wall and transudation through the gut serosa. Lymph “sweating” from the liver surface is the mechanism in X. [Slide 19]
17	**D**	HARD	The lecture names the liver as the only organ with glucose-6-phosphatase, so only it can liberate glucose from glycogen into the circulation. [Slide 43]
18	**B**	HARD	Circulating ammonia comes primarily from the colon and kidneys; the liver is the only organ with the complete urea cycle, and ammonia is toxic to the CNS because it crosses the blood–brain barrier freely. [Slide 40]
19	**C**	HARD	Sinusoids → space of Disse → space of Mall / portal lymphatics → hepatic and regional nodes → coeliac/para-aortic pathways → cisterna chyli → thoracic duct → left venous angle. [Slides 31–33]
20	**A**	HARD	The liver produces thrombopoietin, which promotes thrombocyte production, and catabolises hormones including oestrogen. [Slides 44, 45]